FIE401 - Assignment 3
Does financial literacy cause participation in the stock market?

Darya Yuferova

Overview

Individuals have become increasingly active in financial markets. This increase in the market participation by individuals has been accompanied or even promoted by the advent of new financial products and services. However, some of these products are complex and difficult to grasp, especially for financially unsophisticated investors.

At the same time, market liberalization and structural reforms to social security and pensions have caused an ongoing shift in decision-making responsibility away from the government and employers and towards private individuals. Thus, individuals have to take more responsibility for their own financial well-being. This process is not universally seen as only positive since not all individual might have the skill and knowledge to do so.

This started a long string of research on the stock market participation of private individuals. In this exercise, you investigate one of the first questions asked in this literature: are individuals that have more finance related knowledge (a concept often referred to as financial literacy) more likely to participate in the stock market?

Formalities

This assignment will be handed out on 24.03.2026 at 12:00 and has to be submitted no later than the **31.03.2026 at 09:00**. Submit your commented coding file (file extension: ".R") and a pdf file (file extension: ".pdf") including the numerical results and answers to questions posted.

Please comment your code shortly so that a reader can reconstruct your thinking. You do not need to explain the used functions. You do not need to describe your coding in the pdf file. Please keep your answers very brief. **Code without comments will automatically result in a "Fail" grade.**

On Canvas, you will find a guideline of how to make good tables. Note that in the last lecture of this course, we will cover the subject of how to present research in detail.

After submission, the assignments will be randomly redistributed. Read your peer's work carefully and comment within Canvas no later than **07.04.2026 at 09:00**. Your comments are meant to improve the work of your peers, so use constructive feedback. See peer-review guidelines for details. **Insufficient quality of peer-review or no peer-review will result in a "Fail" grade, regardless of the quality of your own analysis.**

In order to make it easy for us to allocate individual assignment submission as well as peer review to groups, please follow following file name structure. **For the assignment submission: "Assignment3_group_XX.R/pdf".** "XX" indicates the group number that submitted the assignment. **Please submit the peer review as a comment on CANVAS.**

Please do not include any personal information such as name, student number, etc. in the submitted files.

Note that in order to complete the assignment, you need to submit your own work as well as complete the peer review. Both aspects will be reviewed.

Work together in groups of 3-4 people.

Submit your assignment even if you do not finish all tasks.

Note that some concepts are introduced for first time in this assignment. Please use online resources actively.

We touched on almost all functions you have to use to solve this assignment. If you are unsure how to use a function, either use R's own documentation (type ?command) or use www.stackoverflow.com. We encourage you to code this assignment yourselves, and not to use purpose-made solutions or packages as provided on the internet.

Before starting to code, try to separate the task into smaller pieces.

Data provided for this assignment

The *Finlit.csv* file provides a dataset allowing you to explore the aforementioned question. In specific, you may find following variables.

- Main dependent variable
  - dstocks_mut: dummy for stock market participation
- Main independent variable
  - indexlit1_new1: advanced literacy index
- Controls
  - age: age in number of years
  - edu2: dummy for pre-vocational education
  - edu3: dummy for intermediate vocational education
  - edu4: dummy for secondary pre-university education
  - edu5: dummy for higher vocational education
  - edu6: dummy for university education
  - male: dummy for gender
  - partner: married or not
  - numkids: number of kids
  - retired: retired or not
  - dum_selfempl: dummy for self-employment
  - lincome: log(houselhold income)
  - tot_non_equity_wealth_cat: wealth quartile
  - indexlit2_new1: basic literacy index
  - b2: economics education
  - b3: daily use of economics
- Potential instruments
  - f10: sibling's financial situation
  - f15: parent's financial knowledge
- Included variables not relevant for current analysis
  - edu1: dummy for primary education
  - b1: self-assessed literacy

Tasks

You estimate following model using a liner probability model:

$$dstocks\_mut_i = \alpha_0 + \beta_1 \times indexlit1\_new1_i + \beta_X \times Controls_i + \varepsilon_i$$

Controls indicates all control variables from the list of controls.

In specific, execute following tasks:

- Task 1: Construct summary statistics to familiarize yourself with the data. If necessary make some adjustments to the data.
- Task 2: Use a *stargazer* table to report following three estimated models
  - Naive OLS regression
  - First stage of the IV regression
  - Second stage of the IV regression estimated in one go, so that standard errors are correct
  - Interpret your results. What does financial literacy mean for stock market participation, statistically and economically? Is there a difference between OLS and IV?
  - Make sure that you use heteroskedasticity robust standard errors.
- Task 3: Assess the instruments
  - Are instruments relevant?
  - Are instruments exogenous?
- Task 4: You have replicated tables from Van Rooij et al. (2011). You can find the paper here: https://doi.org/10.1016/j.jfineco.2011.03.006. Please note that in order to download the paper you need to be connected to NHH network (can be achieved via Virtual Desktop). In particular, you have replicated columns in Table 7, 8A and 8B. Carefully read chapter 5 (start page 461 at the second paragraph beginning with "In Table 7,...") of the paper and answer following questions:
  - What reasons do the authors name for using an Instrumental Variable approach?
  - How do the authors abstract the estimates obtained using OLS and IV?
  - How do the authors discuss relevancy? (F-statistic)
  - After reading the authors explanations, are you convinced that both instruments are valid? (Exogeneity)

References

Van Rooij, M., Lusardi, A., Alessie, R., 2011. Financial Literacy and Stock Market Participation, Journal of Financial Economics, 101, pp. 449-472

```r
##############################################
#                                            #
#  Suggested code solution for Assignment 3  #
#                                            #
##############################################


# clear workspace
rm(list = ls())

# load relevant packages
require(car)
require(AER)
require(stargazer)

# load the data
finlit <- read.csv("C:/Users/s13163/Dropbox/FIE401/data/data_assignments/Finlit.csv",
                   sep =",")

# summary stats
summary(finlit)
```

```
##      f10                f15                 b1                 b2
##  Length:1508        Length:1508        Length:1508        Length:1508
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##       b3                 age               numkids         partner
##  Length:1508        Length:1508        Min.   :0.0000   Length:1508
##  Class :character   Class :character   1st Qu.:0.0000   Class :character
##  Mode  :character   Mode  :character   Median :0.0000   Mode  :character
##                                        Mean   :0.7109
##                                        3rd Qu.:2.0000
##                                        Max.   :7.0000
##
##   dum_selfempl     tot_non_equity_wealth_cat    lincome
##  Min.   :0.00000   Min.   :1.000             Min.   : 7.244
##  1st Qu.:0.00000   1st Qu.:2.000             1st Qu.: 9.720
##  Median :0.00000   Median :3.000             Median :10.055
##  Mean   :0.05214   Mean   :2.738             Mean   :10.052
##  3rd Qu.:0.00000   3rd Qu.:4.000             3rd Qu.:10.387
##  Max.   :1.00000   Max.   :4.000             Max.   :13.828
##  NA's   :319       NA's   :392               NA's   :112
##      male           indexlit1_new1    indexlit2_new1        edu1
##  Length:1508        Min.   :-2.9203   Min.   :-4.6798   Min.   :0.00000
##  Class :character   1st Qu.:-0.4447   1st Qu.:-0.1460   1st Qu.:0.00000
##  Mode  :character   Median : 0.4492   Median : 0.4904   Median :0.00000
##                     Mean   : 0.0000   Mean   : 0.0000   Mean   :0.04446
##                     3rd Qu.: 0.7592   3rd Qu.: 0.7918   3rd Qu.:0.00000
##                     Max.   : 1.0240   Max.   : 0.7918   Max.   :1.00000
##                                                         NA's   :1
```

```
##       edu2             edu3             edu4             edu5       
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000  
##  Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000  
##  Mean   :0.2289   Mean   :0.1951   Mean   :0.1374   Mean   :0.2634  
##  3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000  
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000  
##  NA's   :1        NA's   :1        NA's   :1        NA's   :1       
##       edu6           retired         dstocks_mut    
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000  
##  Median :0.0000   Median :0.0000   Median :0.0000  
##  Mean   :0.1307   Mean   :0.2003   Mean   :0.2868  
##  3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000  
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000  
##  NA's   :1                         NA's   :319     
```

```
#drop incomplete observations
finlit<-na.omit(finlit)

# Task 2.1: naive OLS
fit_ols <- lm(dstocks_mut ~
                # main independent
                + indexlit1_new1
                # controls
                + age + edu3 + edu4 + edu5 + edu6 + male
                + partner + numkids + retired + lincome + tot_non_equity_wealth_cat
                + indexlit2_new1 + b2 + b3
                # data
                , data = finlit)

# robust standard errors
se_fit_ols <- coeftest(fit_ols,vcov=hccm)[,2]


# Task 2.2: First stage regression
fit_1stage <- lm(indexlit1_new1 ~
                   # instruments
                   (f10=="worse") + ( f10=="better") +
                   (f15=="intermediate or high") + (f15=="dont know")
                   # controls
                   + age + edu3 + edu4 + edu5 + edu6 + male
                   + partner + numkids + retired + lincome + tot_non_equity_wealth_cat
                   + indexlit2_new1 + b2 + b3
                   # data
                   , data = finlit)

# robust standard errors
se_fit_1stage <- coeftest(fit_1stage,vcov=hccm)[,2]

# Check instruments relevance
# F-statistic for weak instruments
# Note that the backslashes "\" are needed as escape
# characters as " otherwise refers to the beginning/end of a string
```

```r
# If you are unsure what the exact variable names are, look at summary(fit_1stage)

myH0 <- c("f10 == \"worse\"TRUE",
          "f10 == \"better\"TRUE",
          "f15 == \"intermediate or high\"TRUE",
          "f15 == \"dont know\"TRUE")
# robust SE
linearHypothesis(fit_1stage, myH0, vcov = hccm)
```

```
##
## Linear hypothesis test:
## f10 == "worse"TRUE = 0
## f10 == "better"TRUE = 0
## f15 == "intermediate or high"TRUE = 0
## f15 == "dont know"TRUE = 0
##
## Model 1: restricted model
## Model 2: indexlit1_new1 ~ (f10 == "worse") + (f10 == "better") + (f15 ==
##     "intermediate or high") + (f15 == "dont know") + age + edu3 +
##     edu4 + edu5 + edu6 + male + partner + numkids + retired +
##     lincome + tot_non_equity_wealth_cat + indexlit2_new1 + b2 +
##     b3
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df      F    Pr(>F)
## 1   1093
## 2   1089  4 9.1861 2.668e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Note that instruments are not very strong (F stat of 9.18).
# Remember rule of thumb: F statistic should be greater than 10.

# Task 2.3: IV estimation
fit_iv <- ivreg(dstocks_mut ~
                  # main independent variable
                  indexlit1_new1
                  # controls
                  + age + edu3 + edu4 + edu5 + edu6 + male
                  + partner + numkids + retired + lincome + tot_non_equity_wealth_cat
                  + indexlit2_new1 + b2 + b3
                  |
                  # instruments
                  (f10=="worse") + ( f10=="better") +
                  (f15=="intermediate or high") + (f15=="dont know")
                  # controls
                  + age + edu3 + edu4 + edu5 + edu6 + male
                  + partner + numkids + retired + lincome + tot_non_equity_wealth_cat
                  + indexlit2_new1 + b2 + b3
                  # data
                  , data = finlit)

# robust standard errors
```

```r
se_fit_iv <- coeftest(fit_iv,vcov=vcovHC(fit_iv))[,2]

# Check instruments exogeneity
# J test
finlit$res<-residuals(fit_iv)
exog<-lm(res~
           # instruments
           (f10=="worse") + ( f10=="better") +
           (f15=="intermediate or high") + (f15=="dont know")
           # controls
           + age + edu3 + edu4 + edu5 + edu6 + male
           + partner + numkids + retired + lincome + tot_non_equity_wealth_cat
           + indexlit2_new1 + b2 + b3
           # data
           , data=finlit)

myH0 <- c("f10 == \"worse\"TRUE",
          "f10 == \"better\"TRUE",
          "f15 == \"intermediate or high\"TRUE",
          "f15 == \"dont know\"TRUE")

Fstat<-linearHypothesis(exog, myH0)$F[2]
print(Fstat)
```

```
## [1] 1.361349
```

```r
testval<-4*Fstat;
print(testval)
```

```
## [1] 5.445395
```

```r
pval<-1-pchisq(testval,3);
print(pval)
```

```
## [1] 0.1419413
```

```r
# We cannot reject H0: all instruments are exogenous
# BUT:J-test is a weak test, so we need to have a good story as well.
# Personally, I do not think the instruments are exogenous.
# Parental financial knowledge will correlate with parental wealth and their stock holdings,
# both of which clearly affect stock holding of the children,
# through spill-over or endowment f.ex., positively.
# Siblings' financial situation is reported by the objects of the study,
# rather than by the sibling him/herself.
# If I view myself as having stock market related knowledge,
# I might be inclined to answer that my siblings are worse of than me.
# Note, that this is just a hypothesis.

# stargazer output
stargazer(list(fit_ols, fit_1stage, fit_iv),
          se = list(se_fit_ols, se_fit_1stage, se_fit_iv),
          type="text", keep.stat=c("n"), report=('vc*t'))
```

```
##
## =====================================================================
##                                         Dependent variable:
```

```
##                                  ----------------------------------------
##                                  dstocks_mut indexlit1_new1 dstocks_mut
##                                      OLS          OLS       instrumental
##                                                               variable
##                                      (1)          (2)           (3)
## ---------------------------------------------------------------------------
## indexlit1_new1                    0.076***                   0.194**
##                                   t = 6.072                  t = 2.270
##
## f10 == "worse"                                 0.310***
##                                                t = 4.094
##
## f10 == "better"                                  0.116
##                                                t = 1.627
##
## f15 == "intermediate or high"                  -0.211***
##                                                t = -2.729
##
## f15 == "dont know"                             -0.568***
##                                                t = -4.256
##
## age> 60                             0.086        0.079         0.080
##                                   t = 1.450    t = 0.580     t = 1.308
##
## age30s                              0.010        -0.093        0.023
##                                   t = 0.203    t = -0.839    t = 0.453
##
## age40s                              0.041        -0.123        0.058
##                                   t = 0.844    t = -1.127    t = 1.088
##
## age50s                              0.030        -0.011        0.033
##                                   t = 0.592    t = -0.096    t = 0.624
##
## edu3                                0.026        0.049         0.019
##                                   t = 0.709    t = 0.570     t = 0.489
##
## edu4                                0.021       0.224***       -0.005
##                                   t = 0.493    t = 2.594     t = -0.111
##
## edu5                               0.065*       0.197***       0.039
##                                   t = 1.726    t = 2.733     t = 0.910
##
## edu6                               0.097**      0.351***       0.054
##                                   t = 2.049    t = 4.379     t = 0.912
##
## malemale                           0.058**      0.312***       0.023
##                                   t = 2.074    t = 5.358     t = 0.628
##
## partnerYes                         -0.029       -0.126*        -0.013
##                                  t = -0.912    t = -1.932    t = -0.379
##
## numkids                            0.0005       -0.024         0.004
##                                   t = 0.032    t = -0.839    t = 0.227
##
```

```
## retired                      -0.022      -0.015      -0.022
##                           t = -0.428  t = -0.153  t = -0.421
##
## lincome                      0.080***     0.009      0.075***
##                            t = 3.025   t = 0.168   t = 2.728
##
## tot_non_equity_wealth_cat    0.051***    0.130***     0.035*
##                            t = 3.689   t = 4.357   t = 1.841
##
## indexlit2_new1                0.014      0.270***    -0.020
##                            t = 1.368   t = 9.627   t = -0.742
##
## b2hardly at all              -0.028     -0.495***     0.030
##                           t = -0.572  t = -5.680   t = 0.466
##
## b2little                     -0.010     -0.223***     0.015
##                           t = -0.217  t = -3.063   t = 0.303
##
## b2some                       -0.019     -0.161***    -0.002
##                           t = -0.450  t = -2.590  t = -0.052
##
## b3hardly at all              -0.055      -0.293**    -0.023
##                           t = -0.869  t = -2.112  t = -0.328
##
## b3little                    -0.142***   -0.308***    -0.107*
##                           t = -2.944  t = -3.768  t = -1.922
##
## b3some                       -0.103**    -0.153**    -0.086*
##                           t = -2.328  t = -2.103  t = -1.830
##
## Constant                     -0.641**     -0.104     -0.574**
##                           t = -2.519  t = -0.197  t = -2.172
##
## ---------------------------------------------------------------------
## Observations                  1,115       1,115        1,115
## =====================================================================
## Note:                                *p<0.1; **p<0.05; ***p<0.01
```